Answers and talk notes

Understand Inequalities and Their Solutions

What your student is learning

Tonight we TEST values in an inequality — including the boundary number — and then graph what we found.

Answers

Ask your student to explain before you look.

1. **A.** Yes, because 5 > 3 is true. Yes, because 5 > 3 is true.
2. **B.** Yes, because 4 ≥ 4 is true. Yes, because 4 ≥ 4 is true when the two numbers are equal.
3. **C.** No, because 4 < 4 is false. No, because 4 < 4 is false.
4. **D.** 9. 9 is a solution of x ≤ 9, because 9 ≤ 9 is true.
5. **C.** Many. Many, because every number greater than 2 makes it true.

Talk about it

Was the boundary number a solution? How does that tell you which circle to draw?

Home mission

Find a real coupon or receipt. Say: "Over $20 gets a free drink." Is 20 > 20 true? Test $21 and $25 the same way. What does that tell you about the circle at 20?

When they are stuck, ask

- Put that number in. Is the statement true or false?
- What happens when you test the boundary number itself?
- Does the symbol point toward bigger numbers or smaller ones?

Try not to say

- An inequality has one answer, find it.
- x ≥ 6 gets an open circle, same as x > 6.
- Shade the way the original problem read, before solving.